

7. Which of the following is true about vector quantities?
- Two perpendicular vectors can give zero resultant
 - If resultant of two vectors is zero, then the vectors are equal and opposite
 - Energy is an example of vector quantities
 - None of the above
8. A vector can be represented by a letter (symbol) with an arrow over its head, this representation refers to:
- Graphical method
 - geometric method
 - algebraic method
 - A and B
9. The magnitude and direction of a force which has x-component 30N and y-component of 40N is
- 70N, 45° north of east
 - 10N South East
 - 50N, $\tan^{-1}(4/3)$ North East
 - 50N North West
10. A girl throws a ball upwards, giving it an initial speed $u = 10 \text{ m/s}$. Neglect air resistance. The time taken to return to the girl's hand and its velocity then, is:
- 2sec and -10m/s
 - 1sec and -20m/s
 - 5sec and 15m/s
 - 4sec and -12m/s.
11. For a vector $A = 2i + 3j + 2k$ and $B = 2i + 10j + 2k$, the y-component of the resultant of A and B is:
- 13
 - 4
 - 12
 - 5
12. The trajectory of a projectile is:
- Hyperbola
 - circular
 - parabola
 - elliptical
13. The change in position of an object is called _____.
- Displacement
 - Distance
 - speed
 - velocity
14. Time taken by a projectile to reach its maximum height is:
- $v_i \sin \theta / g$
 - $v_i^2 \sin 2\theta / g$
 - $v_i^2 \sin^2 \theta / 2g$
 - $2v_i \sin \theta / g$
15. A track covers 40m in 10s while smoothly slowing down to a final speed of 3m/s. The acceleration and initial velocity are:
- 0.3 and 3
 - 1 and 7
 - 0.2 and 5
 - 0.5 and 3
16. Which one of the following is basic and scalar quantity?
- Volume
 - Time
 - Energy
 - Pressure

III. Give short answer for the following question (4 points)

- Explain the two types of mechanics.
- State the Newton's laws of motion.

IV. Work out (10 points) (3%, 3% and 4% respectively)

19. Let vector $A = 6i + 4j + 8k$ and $B = 9i + 5j + 4k$ then find

A. Magnitude of $2A + B$

B. Unit vector in the direction of A and B

20. A projectile is fired in such a way that its horizontal range is equal to its maximum height.

What is the angle of projection?

21. If a mass of 40kg accelerated from velocity $(4i - 5j + 3k)m/s$ to $(8i + 3j - 5k)m/s$ in 10s. Find

A. The magnitude of force

B. Find a unit vector in the direction of the resultant of force.

Answer sheet

NAME _____

ID.NO. _____ Section _____

I. True/False

II. Multiple choice

1. ☐
2. ☐
3. ☐
4. ☐
5. ☐

6. ☐
7. ☐
8. ☐
9. ☐
10. ☐

11. ☐
12. ☐
13. ☐
14. ☐
15. ☐

16. ☐